DEPARTMENT OF ELECTRONICS &COMMUNICATION
ERING UNIVERSITY COLLEGE OF ENGINEERING (A)

Osmania University, Hyderabad-07

CLASS TEST-II

SUBJECT :SIGNAL AND SYSTEMS

ass: BE (CSE & AIML) **Max Marks: 30**

emester: IV **Duration: 90 Min**

Academic Year:2026-27

Date:

	PART A Answer All Questions (10 M)	Marks	BT	CO
1. a)	Find Fourier Transform of $e^{3t}u(t)$	2	1	3
b)	Find the Fourier transform of the impulse and step signals	2	2	3
c)	Find the ROC of a function $e^{-at}u(t) + e^{-bt}u(-t)$ $for\ a < b.$	2	3	3
d	Find the z-transform and its ROC of δ(n-2).	2	4	5
e	State the relationship between Laplace transform and Fourier transform.	2	1	3
	PART B Answer Any Two Questions (20 M)			
2. a)	State and prove Parseval's theorem and convolution property in DTFT	6	5	5
b	Prove that the autocorrelation function of an energy signal and its energy spectral density are Fourier transform pairs.	4	6	3
3. a)	State and prove initial-value and final-value theorem of the Laplace transform.	6	3	3
b	Find the Laplace transform of $e^{-a\|t\|}$. Plot ROC.	4	3	3
4. a)	State and prove the convolution and scaling properties of z-transform.	5	4	5
b)	Prove that the sequence $x(n) = a^n$ u(n) and $x(n) = -a^n$u(-n-1) have the same X(z) and differ only in ROC. Also plot their ROC	5	5	5

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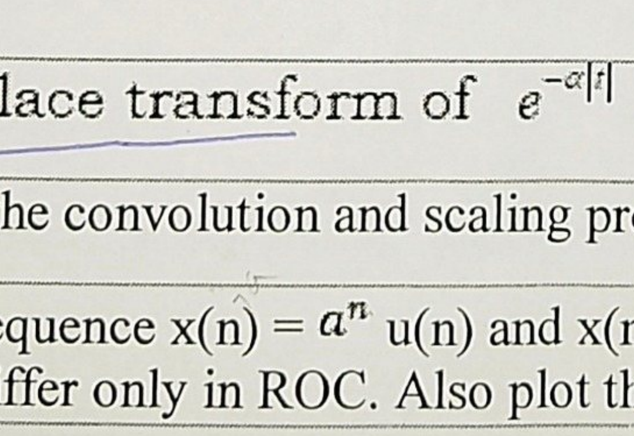
$-\alpha t \quad t>0$

$e \qquad t<0 \qquad \mu(t)$

$x = \begin{cases} 1 & x \neq t \\ 0 & other \end{cases}$

at

e



$\int_0^\infty e^{(\alpha-s)t}\,dt$

$\frac{1}{\alpha-s}\left(e^{0} - e^{-\infty}\right)$

$\int_{-\infty}^{\infty} x(t)e^{-st}\,dt$

$\alpha - s < 0$

$s - \alpha > 0 \quad -(s-a) < 0$

$s > \alpha \qquad s < a$

$s > a$

$\int_0^\infty e^{-at}e^{-st}\,dt + \int$

$T = T, ZT,$

$Lapt.$